

imbalance, and the VWAP price over the period is used as a proxy for the average execution price for the order and is used to compute arrival cost.

Order Data—broker-dealers and vendors (including OMS and EMS companies) have large investor order databases. These actual order sizes and execution prices coupled with the stock specific information serve as input data for the model.

Customer Order Data—customers maintain their own inventory of trades and orders. Clients have access to the full order size and actual execution prices. These data points are used for input in to the model.

ADV = average daily trading volume. This metric is computed based on total market volume during exchange hours over a specified historical period such as a 20 or 30 day period. There has been much discussion regarding the appropriate time period to measure ADV and how the average should be calculated. In Chapter 7, Advanced Algorithmic Forecasting Techniques, we discuss many of the metrics being used and how an analyst can best determine the historical ADV measure for their needs.

V* = actual market volume over the trading period. If the order was traded in the market over the period from 9:30 a.m. to 4:00 p.m. this measure is the actual volume on the day, but if the order was traded over the period from 10:00 a.m. to 2:00 p.m. then this statistic is measured as total market volume during the time the order was traded in the market, from 10:00 a.m. through 2:00 p.m.

σ = annualized volatility expressed as a decimal (e.g., 0.20 and not 20% or 20). It is computed as the standard deviation of log price returns (close-to-close) over the previous 20 or 30 days. In Chapter 6, Price Volatility, we discuss various different volatility forecasting methods used in the industry. The Journal of Trading article titled “Intraday Volatility Models: Methods to Improve Real-Time Forecasts” Kissell (2012) presents techniques on how analysts can develop real-time volatility forecasts to help improve trading decisions and algorithmic trading performance.

Percentage of Volume = computed as the market imbalance or customer order size divided by the actual market volume that traded in the market during the trading period. That is:

$$POV = \frac{Q}{V^*}$$

Arrival Cost = the difference between the execution price of the order and the arrival price of the order, e.g., the mid-point of the bid-ask